

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India **09/2021 – 01/2025 (3.5 Years)**

- Contributed to system validation coverage by reviewing **system requirements** and architecture designs, identifying testability gaps for automotive camera systems and mechatronic ECUs.
- Designed and documented **test specifications** to meet functional and performance requirements, ensuring traceability to system standards throughout the project lifecycle.
- Executed system testing, including **stress and robustness testing**, on automotive software features in bench environments, reporting on detected defects and overall execution status.
- Developed **Python scripts** for automating data validation and reporting tasks, reducing manual effort by **30%** for test data preparation.
- Supported troubleshooting and root cause analysis for system issues identified during validation, contributing to the validation of fixes and ensuring **system reliability**.
- Collaborated with cross-functional teams to define test environment requirements, facilitating the setup and maintenance of necessary hardware-in-the-loop (HIL) and vehicle testing setups.

SKILLS

Core Technologies – Python, C++, C, C#, .NET, Java, Spring Boot, REST APIs, SQL

Testing & Validation – System Validation, Functional Requirements, Performance Requirements, Test Specification, Test Environment, Bench Testing, HIL, Vehicle Testing, Stress Testing, Robustness Testing, Root Cause Analysis, Defect Reporting, Traceability

Automotive Systems – ADAS, Automotive Camera Systems, Mechatronic ECU Systems, Automotive Software Features

Development Tools – Git, Jenkins, Jira, Confluence, Docker

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Linux

Data Analysis & BI – Power BI, DAX, Tableau, R, pandas, NumPy, scikit-learn

Methodologies – Agile, Scrum, Systems Development Life Cycle

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin **02/2026**

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India **01/2021**

PROJECTS

Automated Test Case Generation Framework

- Developed a Python-based framework to automatically generate test cases from system requirements documentation, increasing test coverage by **20%**.

- Integrated the framework with **Jenkins CI/CD pipelines** to enable continuous testing and early defect detection.
- Designed a modular architecture allowing for easy extension to new system components and test types for automotive software features.

Performance Monitoring Dashboard for Enterprise Applications

- Built an interactive dashboard using **Power BI** to visualize real-time performance metrics and system health for enterprise applications.
- Developed complex **DAX measures** to calculate key performance indicators (KPIs) such as response time, error rates, and resource utilization.
- Automated data extraction and transformation using **Power Query**, reducing data refresh times by **40%** and improving data accuracy.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)